protocol.

protocol (UART, I2C, SPI, CAN, CAN-FD) + etc

- features
- Appl^n Areas
- B.D (connectivity)
- Advantages
- Disadvantages
- frames
  - fields

I2C bus events

SPI modes of operation

How many devices/nodes can be interfaced the bus.

Daisy-chain Conf.

which devices you have worked with for respective protocol.

Hurdles you have faced during implementation.

Graduation project / Vector project

- project title
- Appln Areas
- B.D
- flow of the project.
- Limitations

- Implemented code for interfacing (I.e. LCD, ---)
- Integration
- Debugging, Testing on H/w.

→ which ARM based Controller you have worked with & List the features of it.

↳ RAM Size
↳ ROM Size
↳ No. of peripherals.

→ Startup code.
PLL role.
machine cycle / C.C

→ Interrupts.
Watchdog Timer.

CAN (Automobiles Vehicle ~~based~~ Appln)

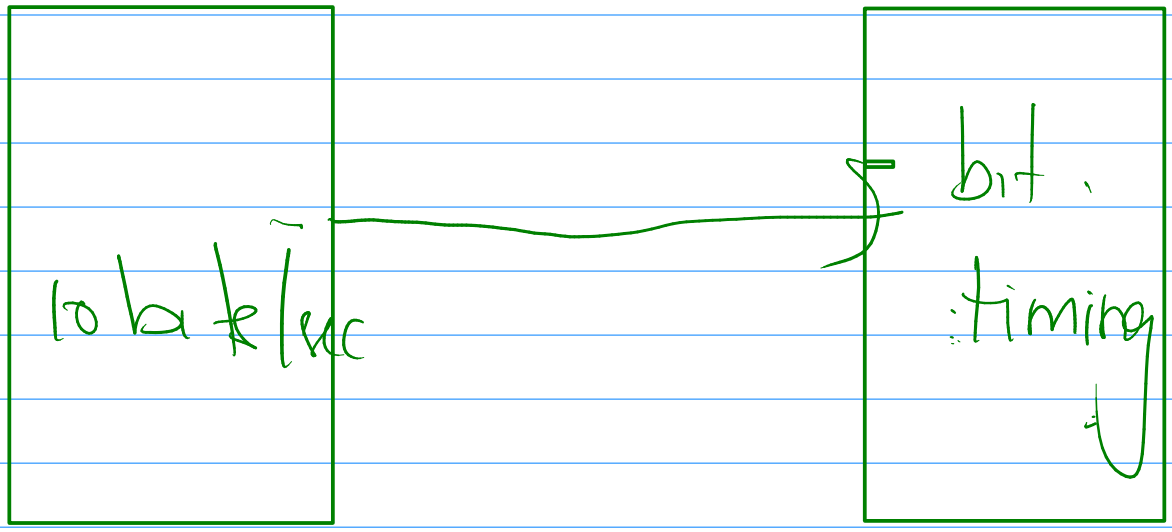
features :-

→ Asynchronous / Serial / Half duplex / wired /

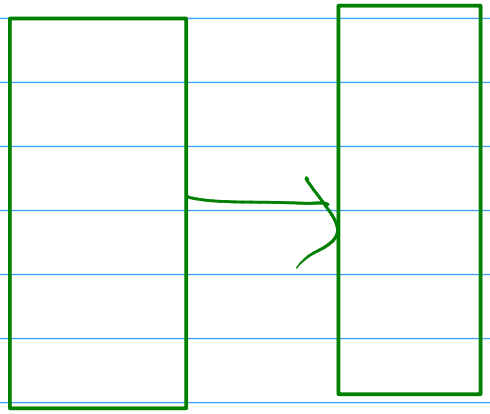
→ Broad cast

→ It support differential voltage.

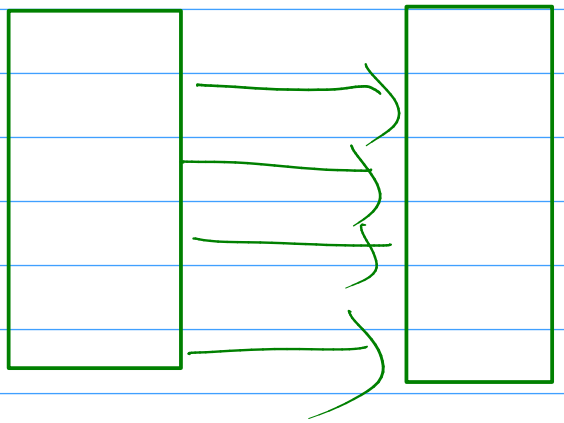
Baudrate / bit timing



Serial



Parallel



→ Half duplex :-

→ Either can transmit / receive at a time

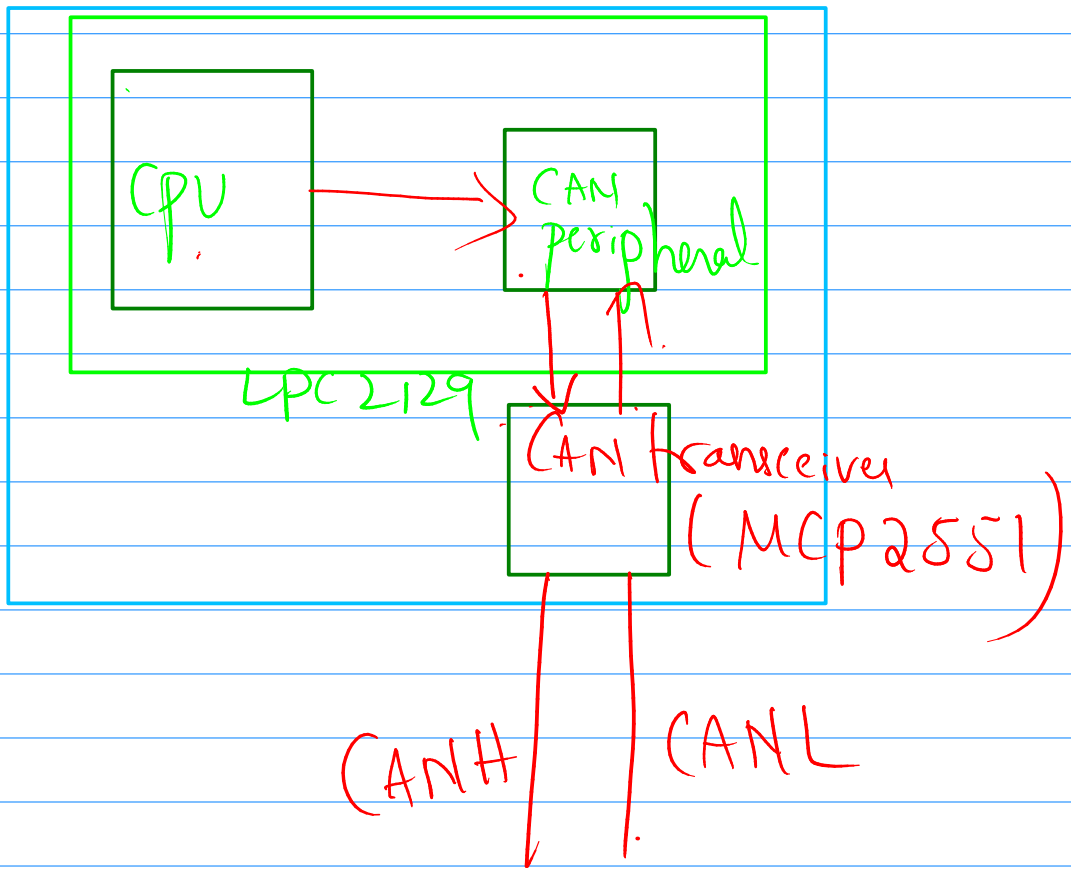
→ Message oriented Comm^n

→ Multi Master

→

Logic 1 — 3.3V | 5V

Logic 0 — 0V



Difference b/w CANH & CANL is determining the logic level.

CAN & CANL difference > 0.8V

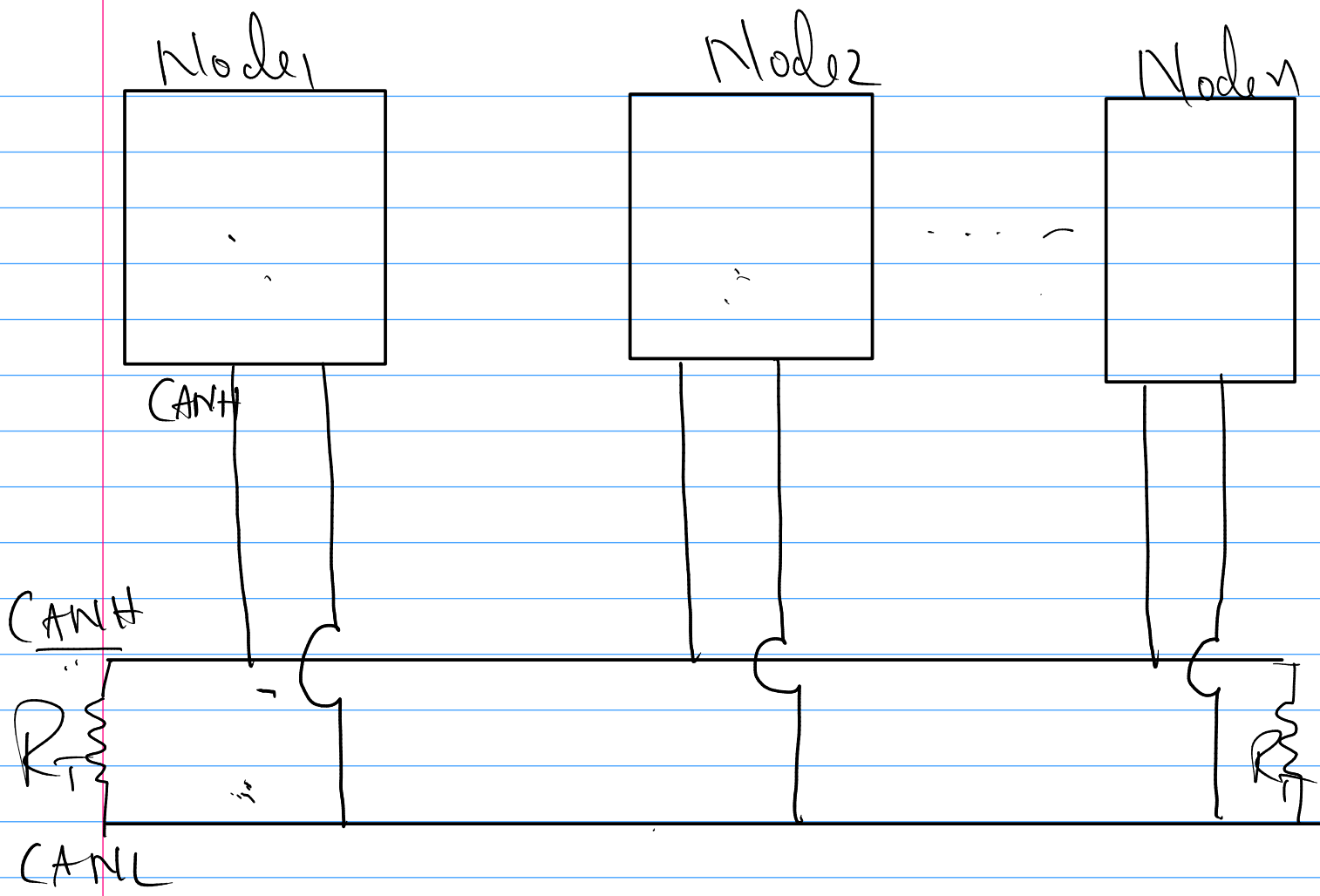
Dominant bit (0)

< 0.5

CANH, CANL → recessive bit (logic 1)

High impedance / floating state / transition state

→ CSMA - CA



Bus arbitration Concept

Node1 0 1 1 1 1 1 1 . . . .

Node2 0 1 1 1 0 1 1 . . . .

Bus level 0 1 1 1 0

when multiple masters are toying for the bus at the same time, then a master can win the bus control if I using wired-AND logic called as Bus arbitration.

→ why CAN ?

1. To reduce wired complexity
2. Adding / Removing a node doesn't need much changes in the n/w.
3. More no. of Error checking methods.

4) priority based Comm^n

5) multi master

Frames

Type Errors

Active-Error

Passive-Error